

Pre-specified validation protocol for sparse_regvelo numerical certification (SAVI)

Protocol version: **v1.0**

Date: **2026-09-14** (precedes all validation runs; earliest harness scripts 2026-09-14, first results 2026-09-15)

Status: time-stamped analysis protocol deposited for review transparency (addresses C4)

Scope: numerical certification and sensitivity analysis of the sparse_regvelo engine on a 74,984-cell Alzheimer's disease microglial atlas (5xFAD/5xFAD;Cd28-cKO mouse model); sensitivity arms on a fixed 5,000-cell subset.

1. Pre-specified pass lines

- **Strict:** per-cell cosine median > 0.99 **AND** relative-L2 median < 0.05
- **Directional** (variable-importance questions only): cosine median > 0.95

2. Metric definitions

- Relative-L2 per cell: $\|v_{\text{test}} - v_{\text{ref}}\| / \max(\|v_{\text{ref}}\|, 10^{-12})$
- Cosine per cell: $(v_{\text{test}} \cdot v_{\text{ref}}) / (\|v_{\text{test}}\| \cdot \|v_{\text{ref}}\|)$; same 10^{-12} floor
- Aggregation: per-cell distributions; median, P99 and max reported separately
- Zero-norm handling: floored at 10^{-12} ; not observed for the non-zero velocity fields in this study

3. Fixed configuration

Production configuration: $K = 10$ segments, cell mini-batch 2048, Picard tolerance $1e-3$, iteration budget 10, adaptive per-gene rate bound $\text{rate_max} = \max(20, 1.2 \times P99.5(\beta, \gamma)) = 107.31$, single precision (FP32), 8 GB device-memory ceiling. Sensitivity arms: fixed 5,000-cell subset (identical cell indices across arms; seed 20260914), $K = 10$, batch 1024, FP32, same adaptive rate-bound protocol.

4. Validation arms

Arm	Definition	Harness script
V1	Stored-output audit vs tightened reference ($K = 20$, $\text{tol} = 1e-6$, 50 iterations); $n = 2,000$ and $5,000$	04_ode_residual*.py
E1	CPU vs CUDA cross-hardware propagation from the faithful adaptive reference, same batch size	03_cpu_gpu_consistency*.py
E2/E2b	Iteration-budget sweep max_iter in $\{10, 20, 30, 50\}$; $K \times \text{tolerance}$ decomposition grid (50 iterations)	09_v1_iteration_sweep_2026-09-15.py
E2c	Per-cell residual tail cross-tabulated against cell-state annotations; Fisher exact + within-state sample permutation across 12 samples	12_e2c_tail_grounding*.py, 12b_e2c_sample_aware_2026-09-16.py
E3 (V7a/V7b)	57 cell-cycle genes removed ($G\ 17,714 \rightarrow 17,657$; $W\ 194,843 \rightarrow 194,216$); two conventions: production moments column-restricted (V7a) vs moments recomputed (V7b)	08_v7_compare_2026-09-15.py, 10_v7a_no_renorm_input_2026-09-15.py
V5	Kinetic rate bound: adaptive (107.31) vs manual 20 vs manual 100, refit on identical data	05_sensitivity*.sh
V6	GRN weight tiers: uniform 1.0 vs mild (1.0/0.75/0.5, -35%) vs aggressive (1.0/0.5/0.25, -57%)	05_sensitivity*.sh, 06_compare_conditions*.py
V8	Segment count K in $\{10, 20, 40\}$ from a common checkpoint (frozen-parameter self-consistency scope)	02_batch_invariance*.py

E4	Fate-level control: kNN graph ($k = 30$, Euclidean) on the shared 10-d latent; CellRank VelocityKernel weights $c_{ij} = \cos(v_i, s_j - s_i)$, $w_{ij} = \text{softmax}(c_{ij}/\sigma_i)$, $\sigma_i = \text{median}_j c_{ij} $ floored at $1e-8$; $T = 0.8 \cdot T_v + 0.2 \cdot T_c$ (Gaussian diffusion kernel, median bandwidth, max-symmetrized); absorption targets $r = 30$ latent-nearest cells per state; exact absorption solve $(I - Q)X = R$ in float64 (SciPy spsolve); row-wise Spearman with ordinal ranks, zero-variance rows excluded	11_e4_fate_control_2026-09-15.py, 11b_e4_extract_labels_2026-09-15.py
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5. Validation disciplines

- **Reference discipline:** parameter-level comparisons use only checkpoints refitted on identical data, never frozen-parameter composites
- **Provenance discipline:** every checkpoint records requested vs actual rate bounds, mode, and SHA-256 fingerprints of β , γ , W and metadata
- **Completeness discipline:** comparison tools require the L3 metadata completion marker, written only after successful completion (rejects half-written outputs)

6. Data and governance

Dataset: 5xFAD/5xFAD;Cd28-cKO mouse microglial atlas (74,984 cells $\times$ 17,714 genes; 12 samples; three genotypes; ages 3/6/8 months), integrated and annotated upstream; controlled-access within the institutional domain. Engine inputs: spliced/unspliced moments (M_s , M_u), 30-d scVI latent truncated to 10 dimensions, regulatory prior W (194,843 edges, 11 transcription factors). Derived validation inputs (subset arrays, GRN prior, manifests) are archived with validation records. No raw data are included in this deposit.

7. Deviations from protocol

None at v1.0. Subsequent deviations and defect resolutions (T0.4 rate-bound override fix; T0.5 metadata repair and re-scoping of V2/V8 as self-consistency checks) are recorded in the validation run.log and CHANGELOG.